



6.1.1 Whose problem is it anyway? ¶

Ponder This 6.1.1.1. What if we solve $Ax = b$ on a computer and the result is an approximate solution $\hat{x}$ due to roundoff error that is incurred. If we don't know x , how do we check that $\hat{x}$ approximates x with a small relative error? Should we check the residual $b - A\hat{x}$?

▼ [Solution](#)

- If

$$\frac{\|b - A\hat{x}\|}{\|b\|}$$

is small, then we cannot necessarily conclude that

$$\frac{\|\hat{x} - x\|}{\|x\|}$$

is small (in other words: that $\hat{x}$ is relatively close to x).

- If

$$\frac{\|b - A\hat{x}\|}{\|b\|}$$

is small, then we *can* conclude that $\hat{x}$ solves a nearby problem, provided we trust whatever routine computes $A\hat{x}$. After all, it solves

$$A\hat{x} = \hat{b}$$

where

$$\frac{\|b - \hat{b}\|}{\|b\|}$$

is small.

So, $\|b - A\hat{x}\|/\|b\|$ being small is a *necessary* condition, but not a *sufficient* condition. If $\|b - A\hat{x}\|/\|b\|$ is small, then $\hat{x}$ is as good an answer as the problem warrants, since a small error in the right-hand side is to be expected either because data inherently has error in it or because in storing the right-hand side the input was inherently rounded.

In the presence of roundoff error, it is hard to determine whether an implementation is correct. Let's examine a few scenarios.

Homework 6.1.1.2. You use some linear system solver and it gives the wrong answer. In other words, you solve $Ax = b$ on a computer, computing $\hat{x}$, and somehow you determine that

$$\|x - \hat{x}\|$$

is large. Which of the following is a possible cause (identify all):

- There is a bug in the code. In other words, the algorithm that is used is sound (gives the right answer in exact arithmetic) but its implementation has an error in it.
- The linear system is ill-conditioned. A small relative error in the right-hand side can amplify into a large relative error in the solution.
- The algorithm you used accumulates a significant roundoff error.
- All is well: $\|\hat{x} - x\|$ is large but the relative error $\|\hat{x} - x\|/\|x\|$ is small.

▼ [Solution](#)

All are possible causes. This week, we will delve into this.